Then, if $n > 1/\delta$, and $z \in D$ we have

$$\begin{aligned} |f_n(z) - f(z)| &= \left| \sum_{k=1}^{n} \int_{(k-1)/n}^{k/n} F(z, k/n) - F(z, s)\, ds \right| \\ &\leq \sum_{k=1}^{n} \int_{(k-1)/n}^{k/n} |F(z, k/n) - F(z, s)|\, ds \\ &< \sum_{k=1}^{n} \frac{\epsilon}{n} \\ &< \epsilon. \end{aligned}$$

This proves the claim, and by Theorem 5.2 we conclude that f is holomorphic in D. As a consequence, f is holomorphic in Ω, as was to be shown.

5.4 Schwarz reflection principle

In real analysis, there are various situations where one wishes to extend a function from a given set to a larger one. Several techniques exist that provide extensions for continuous functions, and more generally for functions with varying degrees of smoothness. Of course, the difficulty of the technique increases as we impose more conditions on the extension.

The situation is very different for holomorphic functions. Not only are these functions indefinitely differentiable in their domain of definition, but they also have additional characteristically rigid properties, which make them difficult to mold. For example, there exist holomorphic functions in a disc which are continuous on the closure of the disc, but which cannot be continued (analytically) into any region larger than the disc. (This phenomenon is discussed in Problem 1.) Another fact we have seen above is that holomorphic functions must be identically zero if they vanish on small open sets (or even, for example, a non-zero line segment).

It turns out that the theory developed in this chapter provides a simple extension phenomenon that is very useful in applications: the Schwarz reflection principle. The proof consists of two parts. First we define the extension, and then check that the resulting function is still holomorphic. We begin with this second point.

Let Ω be an open subset of $\mathbb{C}$ that is symmetric with respect to the real line, that is

$$z \in \Omega \quad \text{if and only if} \quad \overline{z} \in \Omega.$$

Let Ω^+ denote the part of Ω that lies in the upper half-plane and Ω^- that part that lies in the lower half-plane.

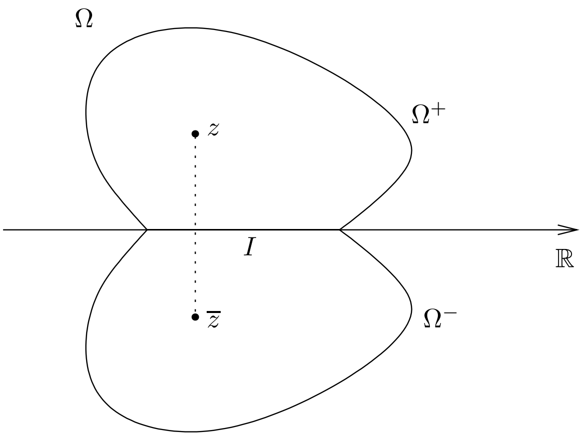

Figure 11. An open set symmetric across the real axis

Also, let $I = \Omega \cap \mathbb{R}$ so that I denotes the interior of that part of the boundary of Ω^+ and Ω^- that lies on the real axis. Then we have

$$\Omega^+ \cup I \cup \Omega^- = \Omega$$

and the only interesting case of the next theorem occurs, of course, when I is non-empty.

Theorem 5.5 (Symmetry principle) *If f^+ and f^- are holomorphic functions in Ω^+ and Ω^- respectively, that extend continuously to I and*

$$f^+(x) = f^-(x) \quad \text{for all } x \in I,$$

then the function f defined on Ω by

$$f(z) = \begin{cases} f^+(z) & \text{if } z \in \Omega^+, \\ f^+(z) = f^-(z) & \text{if } z \in I, \\ f^-(z) & \text{if } z \in \Omega^- \end{cases}$$

is holomorphic on all of Ω.

Proof. One notes first that f is continuous throughout Ω. The only difficulty is to prove that f is holomorphic at points of I. Suppose D is a disc centered at a point on I and entirely contained in Ω. We prove that f is holomorphic in D by Morera's theorem. Suppose T is a triangle in D. If T does not intersect I, then

$$\int_T f(z)\,dz = 0$$

since f is holomorphic in the upper and lower half-discs. Suppose now that one side or vertex of T is contained in I, and the rest of T is in, say, the upper half-disc. If T_ϵ is the triangle obtained from T by slightly raising the edge or vertex which lies on I, we have $\int_{T_\epsilon} f = 0$ since T_ϵ is entirely contained in the upper half-disc (an illustration of the case when an edge lies on I is given in Figure 12(a)). We then let $\epsilon \to 0$, and by continuity we conclude that

$$\int_T f(z)\,dz = 0.$$

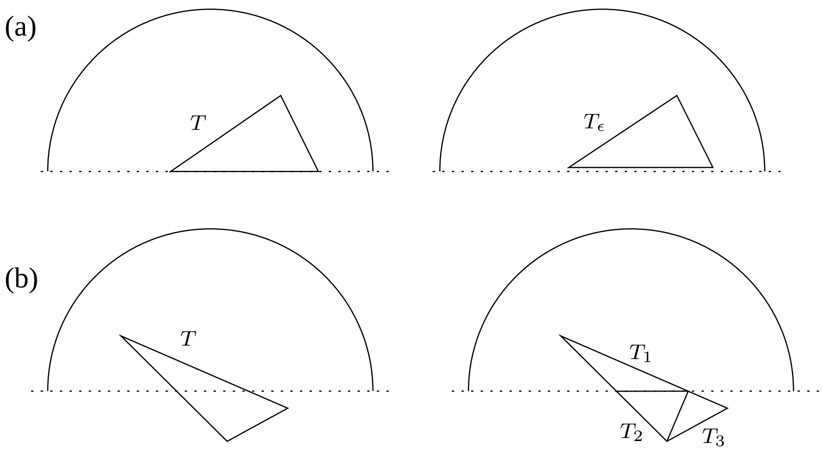

Figure 12. (a) Raising a vertex; (b) splitting a triangle

If the interior of T intersects I, we can reduce the situation to the previous one by writing T as the union of triangles each of which has an edge or vertex on I as shown in Figure 12(b). By Morera's theorem we conclude that f is holomorphic in D, as was to be shown.

We can now state the extension principle, where we use the above notation.

Theorem 5.6 (Schwarz reflection principle) *Suppose that f is a holomorphic function in Ω^+ that extends continuously to I and such that f is real-valued on I. Then there exists a function F holomorphic in all of Ω such that $F = f$ on Ω^+.*

Proof. The idea is simply to define $F(z)$ for $z \in \Omega^-$ by

$$F(z) = \overline{f(\overline{z})}.$$

To prove that F is holomorphic in Ω^- we note that if $z, z_0 \in \Omega^-$, then $\overline{z}, \overline{z_0} \in \Omega^+$ and hence, the power series expansion of f near $\overline{z_0}$ gives

$$f(\overline{z}) = \sum a_n (\overline{z} - \overline{z_0})^n.$$

As a consequence we see that

$$F(z) = \sum \overline{a_n} (z - z_0)^n$$

and F is holomorphic in Ω^-. Since f is real valued on I we have $\overline{f(x)} = f(x)$ whenever $x \in I$ and hence F extends continuously up to I. The proof is complete once we invoke the symmetry principle.

5.5 Runge's approximation theorem

We know by Weierstrass's theorem that any continuous function on a compact interval can be approximated uniformly by polynomials.[4] With this result in mind, one may inquire about similar approximations in complex analysis. More precisely, we ask the following question: what conditions on a compact set $K \subset \mathbb{C}$ guarantee that any function holomorphic in a neighborhood of this set can be approximated uniformly by polynomials on K?

An example of this is provided by power series expansions. We recall that if f is a holomorphic function in a disc D, then it has a power series expansion $f(z) = \sum_{n=0}^{\infty} a_n z^n$ that converges uniformly on every compact set $K \subset D$. By taking partial sums of this series, we conclude that f can be approximated uniformly by polynomials on any compact subset of D.

In general, however, some condition on K must be imposed, as we see by considering the function $f(z) = 1/z$ on the unit circle $K = C$. Indeed, recall that $\int_C f(z)\, dz = 2\pi i$, and if p is any polynomial, then Cauchy's theorem implies $\int_C p(z)\, dz = 0$, and this quickly leads to a contradiction.

[4] A proof may be found in Section 1.8, Chapter 5, of Book I.